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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (AI&ML)

MAJOR PROJECT DOCUMENTATION

MCP-BASED PERSONAL PRODUCTIVITY AGENT

An Intelligent AI Assistant Powered by Model Context Protocol

Under the Guidance of

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ABSTRACT

Managing tasks, schedules, notes, emails, and personal information across multiple applications can be time-consuming and fragmented. **MCP-Based Personal Productivity Agent** addresses this problem by creating a unified AI assistant that can intelligently access and interact with different productivity tools through **Model Context Protocol (MCP)**.

The system uses an **AI agent built with LangGraph** to understand user requests, discover the appropriate MCP tools, and perform actions such as checking schedules, managing tasks, searching notes, or retrieving relevant information. Instead of hard-coding individual integrations, MCP provides a standardized interface that makes the agent's tool usage more flexible, modular, and scalable.

The platform can be developed using **Python, FastMCP, LangGraph, Streamlit**, and synthetic or personal productivity data. Custom MCP servers can provide tools for **calendar management, task tracking, and note searching**, while the conversational interface allows users to interact naturally with the agent.

By combining **Agentic AI, standardized tool integration, and natural-language interaction**, the system demonstrates how MCP can simplify AI-tool communication and enable intelligent assistants that are more **modular, extensible, and adaptable** to different productivity workflows.

Keywords: Model Context Protocol (MCP), Agentic AI, Dynamic Tool Discovery, AI Tool Orchestration,

LangGraph, Natural Language Interaction, Modular AI Architecture, Productivity Automation

INTRODUCTION

Personal productivity has become increasingly dependent on digital tools for managing tasks, schedules, reminders, notes, communication, and information. Although numerous applications are available for these activities, users often need to move between multiple platforms to complete simple workflows. This fragmentation can increase cognitive load, consume time, and make it difficult to maintain a consistent view of pending activities. Artificial intelligence provides an opportunity to simplify these interactions by allowing users to communicate with productivity systems using natural language.

A personal productivity agent can act as an intelligent intermediary between the user and different digital services. Instead of manually navigating individual applications, a user can provide a request such as creating a task, checking a schedule, setting a reminder, or retrieving information. The agent can understand the intent behind the request and select the appropriate tool to accomplish it. For such an approach to work effectively, the AI system requires a reliable mechanism for discovering and interacting with external tools.

The Model Context Protocol (MCP) provides a standardized approach for connecting AI applications with external tools, resources, and services. In an MCP-based architecture, an AI agent can communicate with MCP servers that expose specific capabilities through well-defined interfaces. This separation between the reasoning component and individual tools makes the system easier to extend and maintain. New productivity capabilities can be added by connecting additional MCP-compatible services without redesigning the entire agent.

The proposed MCP-Based Personal Productivity Agent focuses on combining conversational AI with tool-based productivity automation. The system is intended to understand user requests, determine the required action, interact with appropriate MCP tools, and provide a clear response to the user. By bringing multiple productivity operations into one interface, the project aims to provide a more efficient, convenient, and user-friendly approach to personal task management.

LITERATURE SURVEY

Toolformer (2023): LLMs can dynamically use external APIs or tools instead of relying only on internal knowledge. The study shows that a language model can learn when to call a tool, select the required API, provide arguments, and use the returned result in its response. This demonstrates the value of tool-using LLMs for tasks requiring calculation, search, and external information.

Limitation: it does not provide a standardized and modular protocol for connecting many independently developed tools.

Agent Interoperability Survey (2025): This survey studies MCP and other protocols for interoperable agents and compares their approaches to communication, tool or agent discovery, interaction, and security. It highlights the need for standardized interfaces because ad-hoc integrations become difficult to maintain as the number of tools increases. MCP is relevant to productivity agents because it provides a structured way for an AI agent to interact with external tools. **Limitation:** the study is mainly a protocol-level analysis rather than a complete personal productivity implementation.

Tool Preferences in Agentic LLMs (2025): This study examines how agentic LLMs select tools and shows that tool selection can be unreliable and strongly dependent on the descriptions provided for available tools. This is important when an agent has several tools with similar capabilities, because unclear descriptions may lead to incorrect selection. **Limitation:** reliable tool selection cannot be assumed only from the availability of a protocol; clear tool descriptions and suitable agent orchestration are also required.

Literature Gap: The reviewed works establish tool-using LLMs, interoperable agent protocols, and the challenges of tool selection. However, they do not directly provide a unified personal productivity system combining conversational interaction with MCP-based task, calendar, reminder, and note-management tools. The proposed project addresses this gap through a modular MCP-based productivity agent.

PROBLEM STATEMENT

Modern users rely on multiple digital applications to manage their personal, academic, and professional activities. Tasks, calendars, reminders, notes, and other information are frequently maintained in separate systems, requiring users to repeatedly switch between applications and manually perform routine operations. This fragmented workflow can lead to missed tasks, duplicated effort, inefficient time management, and increased cognitive load.

Existing productivity applications generally provide useful individual features, but they do not always offer a unified conversational mechanism for coordinating multiple tools. Conventional automation systems may also require predefined workflows and technical configuration, which can limit flexibility for users who want to express requests naturally. There is therefore a need for an intelligent productivity solution that can understand natural-language instructions and interact with multiple productivity services through a standardized tool interface.

The problem addressed by this project is the development of an MCP-Based Personal Productivity Agent capable of receiving natural-language requests, understanding user intent, selecting suitable external tools, and coordinating productivity-related actions through MCP. The system should provide a unified and user-friendly interaction layer while maintaining a modular architecture that allows additional tools and services to be integrated in the future.